

What Survives Temporal Compression? Frequency-Aware Training of Low-Rate Audio Encoders

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Abstract

Neural audio codecs provide frame-level acoustic representations for spoken language models and general audio generation systems, where temporal rate directly controls sequence length and downstream compute. We study how to train low-rate continuous acoustic encoders that preserve waveform fidelity across speech and general audio while supporting downstream recognition. Naively reducing the frame rate amplifies spectral imbalance: reconstruction losses are dominated by low-frequency structure, while perceptually salient broadband details become harder to preserve under temporal undersampling. We propose a frequency-aware three-stage curriculum combining full-band reconstruction, band-wise refinement with dedicated frequency budgets, and adversarial detail enhancement. Together, these stages enable stable training at 12.5 Hz ($2\times$ compression) while maintaining reconstruction quality and adapter-based ASR performance. A controlled sweep over latent dimension and temporal rate shows that temporal resolution, not channel width alone, is the dominant factor: widening the latent only partially mitigates temporal undersampling within the tested range, and a concatenation baseline suggests that frame-level temporal organization matters beyond raw capacity.

1 Introduction

Many recent spoken language systems separate semantic and acoustic roles in their audio representations. Semantic features from models such as Whisper (Radford et al., 2023) or self-supervised learned encoders (Chen et al., 2022; Hsu et al., 2021; Babu et al., 2021) support automatic speech recognition (ASR) and spoken language understanding (SLU), while codec-style acoustic representations from neural audio codecs such as EnCodec (Défossez et al., 2023) and DAC-VAE (Kumar et al., 2024; Meta Fundamental AI Research, 2023) pro-

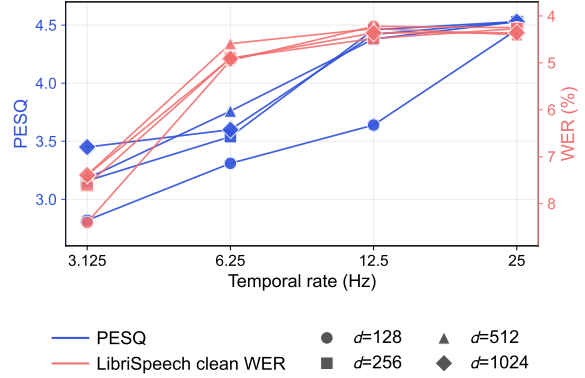


Figure 1: As the rate increases, PESQ improves while LibriSpeech-clean WER decreases, showing that perceptual quality and downstream ASR accuracy follow the same temporal-rate trend across latent dimensions.

vide frame-level acoustic inputs for speech generation, audio modeling, and unified speech language models (Borsos et al., 2023; Wang et al., 2023; Agostinelli et al., 2023; Défossez et al., 2024). Across these systems, the temporal rate of the audio representation directly determines the sequence length processed by downstream models. A 10-second utterance represented at 25 Hz contains 250 frames, whereas reducing the rate to 12.5 Hz halves the sequence length and can substantially reduce the compute of speech LMs, and generative decoders. However, reducing temporal rate is not merely lowering bandwidth. Temporal compression changes which acoustic evidence remains available to downstream models. Fine temporal events and broadband cues, such as fricative energy, consonant bursts, transient attacks, and noisy acoustic textures, become especially vulnerable under aggressive temporal compression (Rosen, 1992; Shannon et al., 1995; Kumar et al., 2024), affecting speech recognition, speaker style preservation, and faithful reconstruction of general audio. Thus, designing low-rate audio tokens requires understanding not only whether downstream metrics drop, but

also what acoustic information is lost under compression.

Our study focuses on reconstruction-capable acoustic encoders for two complementary reasons. First, unlike semantic features, acoustic latents provide the frame-level interface for generative or acoustic-modeling components in many spoken language stacks, so reducing their temporal rate yields direct sequence-length and compute savings rather than serving only as an analysis tool. Second, the encoder–decoder structure provides a closed reconstruction loop for diagnosing compression-induced loss. Semantic encoders, despite their strong task performance, are optimized for invariance and abstraction, making it difficult to attribute degradation to specific acoustic cues or spectral regions (Baevski et al., 2020; Hsu et al., 2021; Chen et al., 2022). A codec-style encoder–decoder, in contrast, lets us measure frequency-localized errors, across speech and general audio, relate them to downstream spoken-language performance, and design objectives that compensate for the observed loss.

In practice, naively reducing the temporal rate of such encoders leads to a characteristic failure mode rooted in *spectral bias* (Rahaman et al., 2019). Standard reconstruction objectives are dominated by energy-rich low frequencies, and temporal compression amplifies this imbalance. The model can preserve coarse low-frequency structure while blurring finer broadband detail, yielding globally coherent but perceptually dull representations. Thus, low-rate acoustic tokens cannot be obtained reliably by simply subsampling an existing encoder; the encoder must be trained under the target temporal bottleneck with objectives that preserve the acoustic evidence most vulnerable to compression.

To address this, we propose a frequency-aware three-stage curriculum for training low-rate continuous acoustic encoders. Full-band spectral reconstruction first establishes a globally coherent latent space. Band-wise refinement then assigns dedicated loss budgets to overlapping low-, mid-, and high-frequency bands so that errors are not dominated by a single spectral region. Finally, adversarial detail enhancement improves fine-grained perceptual structure. Together, these stages enable stable training at the $2\times$ -compressed rate of 12.5 Hz while preserving reconstruction fidelity on speech and general audio, as well as downstream spoken-language performance.

We further ask whether channel width can com-

pensate for temporal undersampling. A common intuition is that increasing the latent dimension d while reducing the temporal rate f , keeping $d \times f$ approximately constant, should preserve information. We test this through a controlled 4×4 sweep over $d \in \{128, 256, 512, 1024\}$ and $f \in \{25, 12.5, 6.25, 3.125\}$ Hz. As shown in Figure 1, our results show that temporal resolution is the dominant axis: along the constant-bandwidth diagonal, the first $2\times$ compression preserves quality with negligible metric change, but further compression causes sharp degradation that additional channel capacity does not fully restore within the tested range. A concatenation baseline that packs adjacent frames without learning separates nominal latent capacity from learned lower-rate organization.

Our contributions are:

- We conduct a controlled study over 16 $\text{dim} \times \text{Hz}$ configurations, showing that temporal rate, rather than latent dimension alone, is the primary determinant of reconstruction quality and ASR performance.
- We propose a frequency-aware three-stage curriculum for training reconstruction-capable acoustic encoders at low temporal rates, enabling high-fidelity 12.5 Hz continuous audio tokens for efficient spoken language modeling.
- We show that temporal compression induces metric-dependent spectral degradation: LF/MF bands absorb most measurable MSE growth, while qualitative high-frequency texture remains perceptually vulnerable. Explicit band-wise loss budgeting improves reconstruction fidelity on both speech and general audio.

2 Related Work

Low-rate representations for speech and audio models. Semantic encoders such as Whisper (Radford et al., 2023) and self-supervised speech models (Chen et al., 2022; Hsu et al., 2021; Babu et al., 2021) provide compact representations for ASR and SLU, but their invariance makes it difficult to attribute compression-induced degradation to specific acoustic cues. Codec latents, in contrast, preserve a reconstruction path and serve as frame-level inputs to audio and speech language models (Borsos et al., 2023; Wang et al.,

2023; Agostinelli et al., 2023; Copet et al., 2023), where sequence length directly affects attention cost. Recent codecs converge on lower temporal rates, often with some form of semantic or task supervision: Mimi (Défossez et al., 2024) operates at 12.5 Hz with RVQ codebooks and semantic-acoustic co-distillation; SemantiCodec (Liu et al., 2024) factorizes semantic and acoustic streams; and X-Codec (Ye et al., 2024) injects pretrained semantic features into the encoder. Qwen3-Omni (Xu et al., 2025) likewise adopts a 12.5 Hz AuT encoder. In contrast, we study how far a purely acoustic, reconstruction-driven encoder with continuous latents can be compressed without external semantic guidance, allowing us to directly measure the waveform- and frequency-level information lost under temporal compression.

Spectral bias in low-rate audio reconstruction.

Neural networks preferentially fit low-frequency components (Rahaman et al., 2019; Xu et al., 2019), and audio reconstruction losses can amplify this tendency because low-frequency content dominates signal energy. Codecs and vocoders mitigate this with multi-resolution spectral losses (Kong et al., 2020; Kumar et al., 2024), multi-band discriminators (Yang et al., 2021), and adversarial training, with vocoders such as HiFi-GAN (Kong et al., 2020) and BigVGAN (Lee et al., 2023) improving high-frequency realism through multi-scale adversarial objectives. These mechanisms improve perceptual quality, but they operate through resolution choices or discriminator architectures rather than explicit loss budgets tied to spectral regions. We argue that temporal compression further exacerbates spectral imbalance, making transient and broadband cues such as fricatives, bursts, and noisy textures harder to preserve; our curriculum therefore allocates per-band loss budgets at the reconstruction-loss level across full-band, band-wise, and adversarial stages.

3 Method

We build a frequency-aware, three-stage curriculum for training a continuous acoustic codec at low temporal rates. The architecture (Sec. 3.1) extends DAC-VAE (Kumar et al., 2024; Meta Fundamental AI Research, 2023) with learnable temporal resampling and a flexible latent dimension, enabling controlled study over (d, f) configurations. Stage 1 (Sec. 3.2) establishes a globally coherent latent space with standard full-band spectral losses.

Stage 2 (Sec. 3.3) then introduces band-wise objectives to counteract metric-dependent spectral imbalance under low-rate training, and Stage 3 (Sec. 3.4) reintroduces adversarial training to refine fine perceptual detail. Finally, Sec. 3.5 describes a concatenation baseline used to probe whether nominal latent bandwidth alone is sufficient in the absence of learned temporal compression. We deliberately scope this study to *continuous* acoustic codecs; discrete (RVQ) codecs introduce additional codebook-rate confounds that are out of scope here.

3.1 Architecture

Our model builds on DAC-VAE (Kumar et al., 2024; Meta Fundamental AI Research, 2023), which encodes 48 kHz audio into 128-dimensional latent vectors at 25 Hz. To enable flexible temporal compression, we introduce two modifications and freeze pretrained low-level features.

Learnable temporal resampling. We insert latent downsample and upsample modules between the encoder, decoder and the variational bottleneck. Each step uses a 1D convolution with kernel size 4 and stride 2, halving the temporal resolution. For 12.5 Hz, one step suffices, for 6.25 Hz, two steps are stacked and for 3.125 Hz, three.

Flexible latent dimension. The encoder output and decoder input projections are adjusted to match the target dimension $d \in \{128, 256, 512, 1024\}$. The VAE mean is bounded by $\tanh(\cdot) \times 10$, and a numerically stable Snake activation (Kumar et al., 2024) prevents overflow.

Partial freezing. The first four encoder blocks and last four decoder blocks are frozen to pretrained weights. Only the final encoder blocks, temporal resampling modules, variational projections, and first decoder block are trained. Unfreezing the full decoder causes training instability and worsens reconstruction artifacts.

3.2 Stage 1: Full-band Spectral Reconstruction

The first stage trains with standard spectral losses: multi-scale mel spectrogram loss (7 scales), multi-resolution STFT loss, KL divergence for VAE regularization, and least-squares GAN with feature matching after warmup. Full formulations are provided in Appendix D.

Config	d	f (Hz)	Speech				Sound		
			SI-SDR↑	PESQ↑	UTMOS↑	MSD↓	SI-SDR↑	PESQ↑	MSD↓
Baseline									
EnCodec (RVQ)	128 (16 RVQ)	150.0	3.51	1.75	1.50	0.92	-1.06	2.38	0.75
SoundStream-16k (RVQ)	512 (8 RVQ)	50.0	-1.00	1.96	2.83	0.76	-18.16	1.60	1.98
ACE-Step VAE	64	25.0	9.60	3.21	3.73	0.44	-1.88	3.03	0.55
MiMi (RVQ)	256 (32 RVQ)	12.5	7.80	3.41	3.74	0.60	-5.85	2.73	1.22
Ours									
128d, 25 Hz	128	25.0	12.22	4.46	3.87	0.16	0.60	4.19	0.25
256d, 12.5 Hz	256	12.5	12.07	4.46	3.88	0.16	0.26	4.20	0.25
512d, 6.25 Hz	512	6.25	5.77	3.76	3.84	0.23	-4.81	3.86	0.33
1024d, 3.125 Hz	1024	3.125	4.28	3.45	3.94	0.29	-28.93	2.54	0.42

Table 1: Reconstruction quality on the same 150-file evaluation set, with 50 speech, 50 sound, and 50 music segments. Baseline rows report public RVQ codecs at their native frame rates, with the dimension column indicating codebook embedding dimension and quantizer count. Ours reports DAC-VAE variants retrained at progressively lower temporal rates while increasing latent width, showing how reconstruction quality changes as temporal resolution is reduced.

3.3 Stage 2: Band-wise Frequency-targeted Refinement

After Stage 1, reconstruction is globally coherent but fine texture appears washed out. The mechanism is the spectral bias of log-magnitude spectral losses: energy-rich low frequencies dominate each optimizer step, so broadband and high-frequency residuals receive a weaker gradient signal even as temporal compression forces each remaining frame to encode more transient detail. Since this is a property of the loss formulation rather than the codec architecture, the same failure mode is expected in any continuous acoustic codec trained with full-band log-magnitude losses, including the SoundStream/DAC/DAC-VAE family. Stage 2 therefore applies the logic of progressive growing (Karras et al., 2018) and the multi-resolution stack of HiFi-GAN (Kong et al., 2020) to the *reconstruction loss* side of the optimization rather than the discriminator side.

Concretely, Stage 2 introduces band-wise objectives over three overlapping bands:

$$\begin{aligned} \text{LF : } 20\text{--}512 \text{ Hz, } \quad \text{MF : } 256\text{--}2048 \text{ Hz,} \\ \text{HF : } 1024\text{--}8192 \text{ Hz.} \end{aligned}$$

The overlap ensures boundary regions are constrained by neighboring bands. Each band term combines: (i) band-wise mel reconstruction, (ii) temporal gradient matching, (iii) frequency-weighted mel loss with slopes $s_{\text{LF}} = 0.5$, $s_{\text{MF}} = 1.0$, $s_{\text{HF}} = 2.0$, and (iv) for HF only, a 2D gradient constraint preserving fine harmonic textures. The HF slope of 2.0 ensures the highest mel bins within the HF band receive twice the loss weight of the lowest, directly counteracting energy-based bias.

3.4 Stage 3: Adversarial Detail Enhancement

Stage 3 reintroduces adversarial training while retaining all band-wise losses. The discriminator combines multi-period (periods 2, 3, 5, 7, 11) and multi-resolution (FFT sizes 2048, 1024, 512) sub-discriminators. Prior band-wise convergence provides a strong initialization that stabilizes GAN training.

3.5 Concatenation Baseline

To isolate the role of frame-level temporal organization, we construct a naive concatenation baseline. Given the original 25 Hz, 128-dimensional DAC-VAE latents $\{z_1, z_2, \dots, z_T\}$, we concatenate adjacent pairs:

$$z'_i = [z_{2i-1}; z_{2i}] \in \mathbb{R}^{256}, \quad i = 1, \dots, T/2$$

producing 256-dimensional vectors at 12.5 Hz. This preserves all original information but discards the temporal ordering within each pair. From a language model’s perspective, the resulting sequence has the same length and dimension as the learned 256d@12.5 Hz representation.

4 Experiments and Results

4.1 Setup

Dataset. We train on a mixed-domain dataset of speech (LibriSpeech (Panayotov et al., 2015)), sound effects (WavCaps (Mei et al., 2023)), and music (MAESTRO v3 (Hawthorne et al., 2019)), totaling approximately 422K segments at 48 kHz.

Metrics. We evaluate with SI-SDR (Le Roux et al., 2019), PESQ (Rix et al., 2001), STOI (Taal et al., 2011), UTMOS (Saeki et al., 2022), mel

spectral distance (MSD), and per-band MSE across three frequency bands (LF: 20–512 Hz, MF: 512–2048 Hz, HF: 2048–8000 Hz).

Training. All 16 grid configurations are trained for 40 epochs with 10K steps per epoch on a single GPU per model. The codec forward pass runs in fp32 with selective AMP on discriminators and gradient checkpointing for memory efficiency. Full training hyperparameters, including optimizer settings, learning rate schedules, and GAN warmup details, are provided in Appendix E.

4.2 Reconstruction Quality

We train DAC-VAE variants at all 16 combinations of $d \in \{128, 256, 512, 1024\}$ and $f \in \{25, 12.5, 6.25, 3.125\}$ Hz. Each configuration modifies the encoder stride and latent projection to natively produce the target rate and dimension.

Table 1 reveals a phase transition rather than a smooth bitrate–quality curve. The first reduction in frame rate, from 25 Hz to 12.5 Hz, is largely absorbed by increasing channel width: the codec still sees acoustic events frequently enough that each latent frame only needs to summarize a short local neighborhood. Below this point, however, widening the latent no longer preserves the missing temporal anchors. The decoder must infer transient timing, excitation changes, and broadband detail from frames that are too far apart, so additional channels mostly improve within-frame description rather than recovering the lost ordering of events.

This pattern is not explained by total bandwidth alone. A higher-bandwidth 12.5 Hz model remains in the high-quality regime, while a same-dimension 6.25 Hz model falls into the low-rate regime. The practical implication is that continuous acoustic codecs have a temporal floor: channel capacity can help near that floor, but once frame spacing becomes too coarse, the limiting factor is no longer how much information each frame can store, but which time-local evidence is sampled at all.

The full 4×4 grid (Table 2) supports the same interpretation away from the diagonal. Moving along the temporal-rate axis changes the qualitative regime of the representation, whereas moving along the dimension axis mostly refines a regime that has already been determined by frame rate. This distinction matters for audio-LM design: shorter sequences are attractive for attention cost, but a lower-rate codec is not simply a shorter version of a higher-rate one. It changes the granularity

d	Temporal rate f (Hz)			
	25.0	12.5	6.25	3.125
128	4.46	3.64	3.31	2.82
256	4.53	4.46	3.54	3.16
512	4.53	4.38	3.76	3.18
1024	4.53	4.42	3.60	3.45

Table 2: Full 4×4 grid: speech PESQ across all $\text{dim} \times \text{Hz}$ combinations. Diagonal entries (bold) hold $d \cdot f \approx 3200$. Quality degrades primarily along the Hz axis (columns) rather than the d axis (rows). Note the matched pair 256d@25 Hz (4.53) $\approx$ 1024d@12.5 Hz (4.42), at $d \cdot f^2 = 160,000$.

at which phonetic transitions, attacks, and texture fluctuations are exposed to the downstream model.

4.3 Three-Stage Curriculum Ablation

Table 3 isolates the curriculum stages in a separate post-hoc compressor setting; the full $\text{dim} \times \text{Hz}$ grid in Table 2 uses natively retrained codec variants. The main lesson is that the three stages serve different roles rather than providing monotonic improvements on every metric. Full-band training establishes a coherent reconstruction but leaves the model with the same frequency imbalance that motivates our method. Band-wise refinement is the most directly aligned with the paper’s central claim: giving LF/MF/HF regions separate budgets makes the representation less dominated by the easiest, highest-energy structure and improves the codec before adversarial training is introduced.

The adversarial stage should be read as a perceptual-detail step, not as a universal fidelity step. It sharpens broadband texture and improves metrics that reward perceptual or spectral plausibility, but it can move the waveform away from sample-level fidelity. This behavior is useful because it clarifies the role of the curriculum: Stage 2 addresses the compression-induced spectral imbalance, while Stage 3 chooses a point on the perception–distortion frontier. Matched-capacity examples show the same convergence pattern, as shown in Figure 2.

4.4 Temporal Rate vs. Latent Dimension

Beyond the qualitative observation that temporal rate dominates, the training curves show that reconstruction quality is not well summarized by the nominal capacity product $d \cdot f$ alone. Although $d \cdot f$ approximates the number of latent dimensions produced per second, configurations with similar values can converge to different quality regimes

Stage	Speech				Sound	
	SI-SDR	PESQ	UTMOS	MSD	SI-SDR	PESQ
Stage 1 (full-band)	4.93	2.82	3.24	2.80	-4.04	2.98
+ Stage 2 (band-wise)	5.44	3.11	3.31	2.29	10.23	2.96
+ Stage 3 (GAN)	3.63	3.27	3.08	2.25	-0.79	3.22

Table 3: Three-stage curriculum ablation for the separate post-hoc compressor setting (256d, 12.5 Hz). Stage 2 (band-wise) provides the largest improvement.

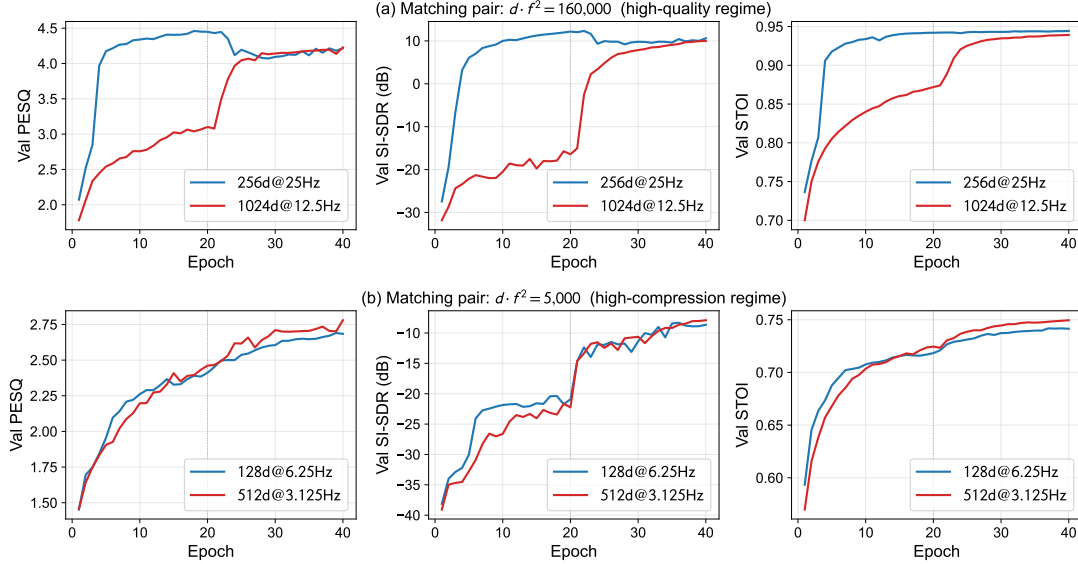


Figure 2: Overlaid training curves for two $d \cdot f^2$ -matched pairs. **Top**: 256d@25 Hz vs 1024d@12.5 Hz ($d \cdot f^2 = 160,000$). **Bottom**: 128d@6.25 Hz vs 512d@3.125 Hz ($d \cdot f^2 = 5,000$). Dotted line: GAN warmup onset (epoch 20). Each pair converges to nearly identical final values across all three perceptual metrics; intermediate epochs may diverge transiently.

when their temporal rates differ. We therefore use $d \cdot f^2$ as a simple empirical descriptor, which better matches the observed trends by giving additional weight to temporal rate.

This pattern is consistent with the structure of audio reconstruction: lowering f not only reduces the number of latent frames, but also increases the waveform interval each frame must summarize. Increasing d adds per-frame capacity, but it does not fully replace the temporal observations lost when f is reduced. The aggregate trend over all grid cells is shown in Figure 3.

Quantitative regression across 16 configurations.

To go beyond the matched examples, we fit log-linear regressions over all retrained cells (Appendix J). The purpose is not to claim a universal law, but to test whether the proposed interpretation survives outside hand-picked pairs. Across speech, sound, and music, the quadratic capacity proxy consistently organizes the grid better than the linear information-rate proxy. This consistency is the important result: the same temporal penalty appears

across domains with different acoustic statistics, suggesting that it is tied to the codec bottleneck rather than to one dataset or metric.

The quadratic scaling is consistent with the following heuristic: halving the frame rate removes one factor of f worth of coherent temporal structure (which the decoder can still reconstruct from neighboring frames), while degrading another factor of f worth of transient detail that *cannot* be inter-frame interpolated. Each remaining frame must therefore encode not just $2 \times$ the information (linear in f) but also bridge the additional transient detail, doubling its effective information load twice, once for absent frames and once for lost transients. Linear doubling of d cannot cover both; quadratic ($4 \times$) roughly can. This is consistent with the band-wise error redistribution documented in §4.4.1, but we present it as an empirical hypothesis pending broader confirmation across codec families.

Practical implication. For audio LMs that tokenize codec latents, dropping f from 25 Hz to 12.5 Hz halves sequence length, but the compen-

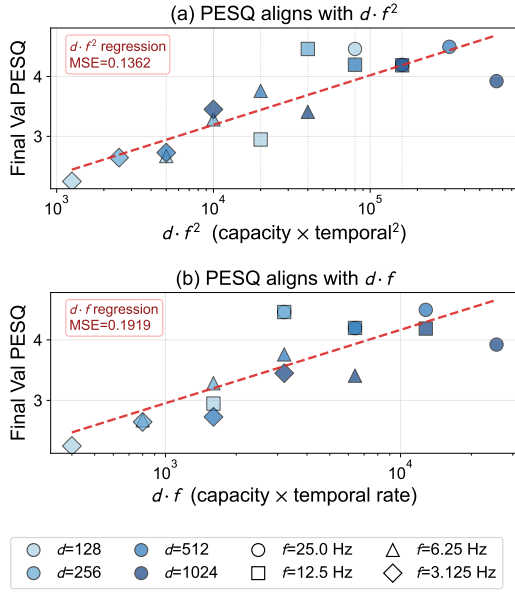


Figure 3: Final PESQ as a function of simple capacity descriptors across all 16 retrained configurations. (a) PESQ plotted against $d \cdot f^2$, which gives additional weight to temporal rate. (b) PESQ plotted against $d \cdot f$, the nominal number of latent dimensions produced per second. Vertical dotted lines indicate configurations with the same descriptor value, and red dashed lines show log-linear regression fits for each x-axis.

sating $4\times$ increase in d per frame means roughly $2\times$ more parameters per sequence, not fewer. The win is context length, not parameter efficiency. Designers weighing this trade-off should factor in the quadratic cost of capacity.

4.4.1 Per-band Spectral Error under Temporal Compression

Semantic information in speech is strongly supported by low- and mid-frequency structure, while perceptual naturalness also depends on high-frequency fidelity such as fricatives, breathiness, and vocal texture. The per-band analysis in Table 5 shows that these two notions of damage are not identical. Under the MSE lens, temporal compression is most visible in LF/MF bands, where formant and voicing structure must remain numerically aligned. HF MSE changes much less, not because high-frequency detail is preserved, but because this band is already noisy and poorly summarized by aggregate energy error.

This distinction matters for the method. A uniform spectral objective can make the codec look acceptable under global losses while hiding two different failure modes: measurable LF/MF drift and perceptual HF smoothing. Stage 2 addresses

Condition	SI-SDR $\uparrow$	PESQ $\uparrow$	UTMOS $\uparrow$	MSD $\downarrow$
Original 25 Hz	14.28	4.28	3.82	0.46
Concat 256d@12.5	14.28	4.28	3.82	0.46
Learned 256d@12.5	12.07	4.46	3.88	0.16

Table 4: Concatenation baseline vs. learned compression (speech). Both expose a 256d@12.5 Hz representation shape, but Concat is a reversible reshape of the vanilla pretrained 25 Hz latents, whereas Learned is a natively retrained 256d@12.5 Hz codec variant.

d	f (Hz)	LF $\downarrow$	MF $\downarrow$	HF $\downarrow$	Total $\downarrow$
128	25.0	0.0129	0.1637	0.3197	0.0901
256	12.5	0.0259	0.1865	0.3075	0.0888
512	6.25	2.3328	0.5276	0.2901	0.1573
1024	3.125	2.5979	0.9478	0.3307	0.2253

Table 5: Per-band spectral MSE across temporal rates (speech, diagonal configs). LF and MF errors increase sharply as temporal rate decreases, while HF MSE remains nearly flat under this metric.

the first by forcing each band to carry its own reconstruction budget, and the HF-specific gradient terms make the second visible to the optimizer even when aggregate HF MSE is insensitive. The result is a more diagnostic training objective, not merely a reweighting trick.

4.4.2 Concatenation Baseline

Table 4 compares the original pretrained DAC-VAE path, a reversible concatenation of adjacent pretrained latents, and our learned lower-rate representation. The Concat row is intentionally not a learned compression method: it simply exposes the native 25 Hz latent stream as a 12.5 Hz, 256-dimensional shape by packing pairs of frames. For reconstruction metrics we unstack it and decode through the original path, so it preserves the pretrained reconstruction by construction. This makes it a control for shape and nominal capacity rather than a competing codec.

The useful takeaway is conservative but important: matching $d \times f$ or packing frames is not enough to characterize a low-rate codec representation. The learned model does not merely store the same information in a shorter sequence; it reorganizes acoustic evidence so the decoder can operate natively at the lower rate. Its metric profile follows the same perception-distortion pattern seen in the curriculum ablation (Blau and Michaeli, 2018), suggesting that learned temporal organization changes which evidence is made accessible, not just how many latent coefficients are available.

Dim	Rate (Hz)	Libri. Clean		Libri. Other	
		WER	CER	WER	CER
128	3.125	8.40	4.54	26.30	16.98
	6.25	4.95	2.30	17.92	11.17
	12.5	4.22	1.92	14.07	8.29
	25.0	4.25	1.90	14.15	8.37
256	3.125	7.60	3.94	23.47	15.02
	6.25	4.90	2.28	16.94	10.31
	12.5	4.48	2.10	15.00	8.83
	25.0	4.27	1.95	14.77	8.58
512	3.125	7.39	3.88	23.17	14.64
	6.25	4.59	2.12	15.77	9.34
	12.5	4.27	1.93	14.93	8.74
	25.0	4.40	2.03	14.71	8.56
1024	3.125	7.39	3.86	23.33	14.68
	6.25	4.91	2.32	16.92	10.13
	12.5	4.36	1.98	14.59	8.54
	25.0	4.36	2.00	15.00	8.76

Table 6: ASR results (WER / CER, %) on LibriSpeech test-clean and test-other for ablations across latent dimension (128–1024) and frame rate (3.125–25 Hz). See Appendix P for bootstrap std values.

4.4.3 Domain Difficulty

We analyze reconstruction quality separately for speech, sound effects, and music. Domain sensitivity is metric-dependent rather than a single total ordering. Sound degrades most strongly in PESQ along the diagonal, while music shows substantial SI-SDR degradation and qualitative loss of harmonic detail at low temporal rates; speech remains comparatively robust in PESQ. Detailed per-domain figures are provided in Appendix K.

Downstream ASR. Table 6 asks whether the compression effects observed in reconstruction also matter once the representation is consumed by a downstream language model. We freeze each acoustic encoder and train a small adapter to project its latents into a frozen Qwen3.5-4B on ~56,000 hours of ASR transcription data (LibriSpeech, GigaSpeech, Voxpopuli, MLS; see Appendix P), probing how much task-relevant temporal evidence remains. This is a strict test for temporal compression: ASR can survive substantial spectral degradation when envelope cues are preserved (Shannon et al., 1995), but temporal structure itself carries linguistic contrasts at multiple time scales.

Representations at 25 Hz and 12.5 Hz form a high-performance regime in which `test-clean` WER stays below 4.5 across every latent dimension, with the spread across configurations falling close to the bootstrap noise floor (Appendix P). The first $2\times$ compression preserves the phonetic and

lexical cues the adapter needs, and extra channel width yields only marginal gains, the adapter is not bottlenecked by per-frame dimensionality in this range.

The behavior changes as the rate drops further. Halving the rate to 6.25 Hz pushes `test-clean` WER up to roughly 4.9 except for the $d=512$ encoder, which holds at 4.59, and `test-other` WER climbs by 1–3 points. At 3.125 Hz the error roughly doubles on both splits, reaching 7.39 `test-clean` WER even for the widest latents. Increasing the latent dimension partially compensates at 6.25 Hz but cannot close the gap to the higher-rate regime, and at 3.125 Hz no choice of dimension recovers the missing temporal evidence. Since ASR mainly needs intelligibility rather than waveform fidelity, this persistent low-rate degradation implies that aggressive temporal compression removes *linguistically* useful information — channel width can store more per frame, but it cannot recover the ordering and localization of short acoustic events such as consonant releases and fricatives. Combined with the reconstruction-side analysis, this points to 12.5 Hz as a practical compression target; pushing below 6.25 Hz creates a temporal undersampling bottleneck that wider latents only partially mitigate.

5 Conclusion

This work shows that reconstruction-capable acoustic encoders can be compressed substantially, but temporal rate is not interchangeable with channel width. Across reconstruction metrics, adapter-based ASR, and the controlled $\text{dim}\times\text{Hz}$ grid, wider latents help some low-rate models but cannot recover undersampled acoustic events. The 25 Hz-to-12.5 Hz reduction is therefore a useful knee point, cutting sequence length while preserving most speech reconstruction and downstream ASR behavior; lower rates, especially 3.125 Hz, degrade fidelity and linguistic evidence. Low-rate failure is also spectral: full-band losses can hide band-specific damage, high-frequency texture remains fragile, and adversarial refinement trades distortion for perceptual detail. The frequency-aware curriculum targets this imbalance with band-wise loss budgets and adversarial detail refinement. Overall, efficient audio tokens should be designed around downstream temporal evidence, not only nominal bitrate or latent dimensionality.

Limitations

Our analysis is intentionally scoped to continuous, reconstruction-capable acoustic encoders. The controlled $d \times f$ grid isolates the role of latent dimension and temporal rate, but does not exhaust the design space of neural codecs, discrete tokenizers, or semantic speech encoders. We also rely mainly on objective reconstruction metrics and adapter-based ASR; future work could add larger-scale listening tests and additional downstream tasks.

Ethical Considerations

Our work improves efficiency of audio language models by reducing sequence lengths. While this broadly benefits accessibility of audio AI, compressed representations could potentially be misused for deepfake generation. We note that our compression operates on already-public codec architectures and does not introduce new synthesis capabilities beyond what existing models already provide.

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Appendix

A Scope of Codec Baseline Comparisons

This work focuses on *continuous, reconstruction-capable acoustic encoders*. The central question is how temporal rate and latent dimension interact when an encoder must still support waveform reconstruction through a decoder. This scope differs from discrete neural codecs and semantic speech encoders, which are important baselines for deployed spoken-language systems but answer a different representation question.

Discrete RVQ tokenizers such as EnCodec/DAC-style codebook systems, Mimi-like low-rate tokenizers, and semantic or semantically enhanced codecs introduce additional design axes: codebook size, number of codebooks, bitrate allocation, residual quantization depth, semantic supervision, and token sampling strategy. These choices are not directly comparable to a continuous vector bottleneck with dimension d and frame rate f . They are useful external systems for end-to-end benchmarking, but they do not isolate the continuous dimension–rate trade-off studied in our 4×4 grid. We therefore treat such systems as complementary codec baselines rather than as controls for the mechanism analyzed here.

Whisper is a supervised ASR encoder, while HuBERT and WavLM are self-supervised speech representation encoders. They are useful reference points for speech understanding, but they are outside the reconstruction-codec setting studied here. These encoders do not provide a native waveform decoder, and their training objectives do not define a reconstruction loop in which full-band, band-wise, or adversarial losses can be applied. As a result, they cannot test whether a low-rate representation jointly preserves acoustic detail for generation and task-relevant evidence for recognition. Applying the proposed GAN or band-wise reconstruction curriculum to these encoders would require adding a separate generative decoder, which would change the object of study.

The ASR downstream-task experiments in this paper should be read under this scope. They test whether continuous acoustic latents remain useful in an adapter-based recognition pipeline, not whether they dominate semantic encoders on ASR alone. Semantic encoders and discrete tokenizers are the right comparison for downstream-only recognition or speech-language-model benchmarking; continuous codec variants are the right com-

parison for the reconstruction-centered question addressed here.

B Model Architecture

Figure 4 shows the full Modified DAC-VAE architecture used for all 4×4 grid configurations. Only the encoder tail (blocks 4–6), VAE projections, decoder head (block 0), and the LatentDown/Up adapters are trained; the rest of the pretrained DAC-VAE (encoder blocks 0–3, decoder blocks 1–4) remains frozen to preserve its acoustic feature extraction and waveform synthesis capabilities.

C Dataset and Evaluation Split

All audio is resampled to 48 kHz mono and segmented into 5.12-second clips (245,760 samples). A 95/5 train/validation split is used with a fixed random seed (42). Evaluation uses 50 randomly sampled files per domain from the validation split.

D Training Objectives

This section defines the reconstruction and adversarial objectives used by the curriculum. Stage 1 uses full-band reconstruction losses to establish a stable codec; Stage 2 adds the band-wise objectives below; Stage 3 keeps the band-wise losses and emphasizes adversarial refinement. The band decomposition and frequency weights are shown in Figure 5.

D.1 Stage 1 Loss Components

Multi-scale mel spectrogram loss. Seven scales with $n_{\text{mels}} \in [5, 10, 20, 40, 80, 160, 320]$, window $w \in [32, \dots, 2048]$. $M_s(\cdot)$ denotes the mel magnitude spectrogram at scale s ; the exponent $p=1$ is a fixed power-law applied for numerical stability before the log:

$$\mathcal{L}_{\text{mel}} = \sum_s \left[\left\| \log_{10}(M_s(\hat{x})^p) - \log_{10}(M_s(x)^p) \right\|_1 + \|M_s(\hat{x}) - M_s(x)\|_1 \right] \quad (1)$$

Multi-resolution STFT loss. Nine FFT sizes (32–8192), spectral convergence + log magnitude L1:

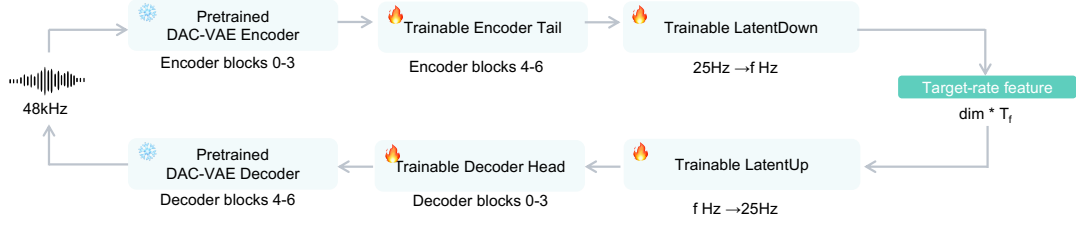


Figure 4: Modified DAC-VAE architecture. We freeze the pretrained low-level encoder and waveform decoder, and train only the encoder tail, temporal resampling modules, VAE projections, and decoder head. LatentDown controls the target temporal rate, while the VAE projection controls the latent dimension d .

Domain	Source	Segments	Hours	Sample Rate
Speech	LibriSpeech train-360	105K	360h	16 kHz
Sound	WavCaps	200K	7500h	48 kHz
Music	MAESTRO v3	117K	200h	48 kHz
Total		422K		

Table 7: Training dataset composition.

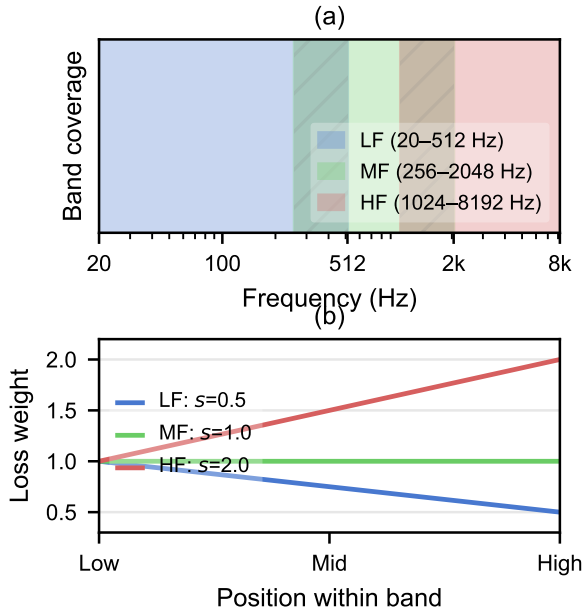


Figure 5: **(a)** Overlapping frequency band decomposition. Hatched regions indicate overlap. **(b)** Frequency weighting slopes: LF ($s = 0.5$), MF ($s = 1.0$), HF ($s = 2.0$).

$$\mathcal{L}_{\text{stft}} = \sum_r \left[\frac{\| |\text{STFT}_r(\hat{x})| - |\text{STFT}_r(x)| \|_F}{\| |\text{STFT}_r(x)| \|_F + \epsilon} + 2 \|\log |\text{STFT}_r(\hat{x})| - \log |\text{STFT}_r(x)|\|_1 \right] \quad (2)$$

GAN losses. Multi-period (periods 2,3,5,7,11) and multi-resolution (FFT 2048,1024,512) discriminators with least-squares loss and feature match-

ing.

Default weights: $\lambda_{kl} = 0.001$, $\lambda_{\text{mel}} = 45$, $\lambda_{\text{stft}} = 10$, $\lambda_{\text{gen}} = 1$, $\lambda_{\text{feat}} = 0.5$. GAN activates after 20-epoch warmup.

D.2 Stage 2 Band-wise Loss Components

Band-wise mel reconstruction. For windows $\mathcal{W} = \{1024, 2048, 4096\}$, cropped mel magnitude per band b :

$$\mathcal{L}_{\text{mel}}^b = \sum_{w \in \mathcal{W}} \left[\left\| \log M_w^b(\hat{x}) - \log M_w^b(x) \right\|_1 + \left\| M_w^b(\hat{x}) - M_w^b(x) \right\|_1 \right] \quad (3)$$

Temporal gradient loss. First and second-order differences of log-mel Z^b :

$$\mathcal{L}_{\text{grad}}^b = \left\| \Delta_t Z^b(\hat{x}) - \Delta_t Z^b(x) \right\|_1 + 0.5 \left\| \Delta_t^2 Z^b(\hat{x}) - \Delta_t^2 Z^b(x) \right\|_1 \quad (4)$$

Frequency-weighted mel loss. Linear per-mel-bin weights $w_i^b = \text{linspace}(1, s_b)_{i \in [1, n_b]}$ inside band b , with slopes $s_{\text{LF}} = 0.5$, $s_{\text{MF}} = 1.0$, $s_{\text{HF}} = 2.0$:

$$\mathcal{L}_{\text{fw}}^b = \sum_{w \in \mathcal{W}} \sum_i w_i^b \cdot \left| M_w^b(\hat{x})_i - M_w^b(x)_i \right|. \quad (5)$$

HF 2D gradient. Joint time-frequency gradient:

$$\mathcal{L}_{\text{2D}}^{\text{HF}} = \left\| \nabla_t Z^{\text{HF}}(\hat{x}) - \nabla_t Z^{\text{HF}}(x) \right\|_1 + 0.5 \left\| \nabla_f Z^{\text{HF}}(\hat{x}) - \nabla_f Z^{\text{HF}}(x) \right\|_1 \quad (6)$$

Composite band terms.

$$\mathcal{L}_{\text{LF}} = 8 \mathcal{L}_{\text{mel}}^{\text{LF}} + 3 \mathcal{L}_{\text{grad}}^{\text{LF}} + 2 \mathcal{L}_{\text{fw}}^{\text{LF}} \quad (7)$$

$$\mathcal{L}_{\text{MF}} = 15 \mathcal{L}_{\text{mel}}^{\text{MF}} + 7 \mathcal{L}_{\text{grad}}^{\text{MF}} + 3 \mathcal{L}_{\text{fw}}^{\text{MF}} \quad (8)$$

$$\mathcal{L}_{\text{HF}} = 15 \mathcal{L}_{\text{mel}}^{\text{HF}} + 8 \mathcal{L}_{\text{grad}}^{\text{HF}} + 5 \mathcal{L}_{\text{2D}}^{\text{HF}} + 4 \mathcal{L}_{\text{fw}}^{\text{HF}} \quad (9)$$

D.3 Stage 3 Adversarial Refinement

Stage 3 retains the Stage 2 band-wise reconstruction terms and continues adversarial training from the refined checkpoint. This stage is intended to sharpen perceptual texture, but as shown in Table 3, it introduces a PESQ/MSD versus SI-SDR/UTMOS trade-off rather than a uniform improvement across metrics.

E Training Hyperparameters

Numerical stability. The codec forward pass is computed entirely in fp32 to prevent overflow in KL divergence computation and Snake activation reciprocals. Mixed-precision (fp16) is applied only to the discriminator forward and backward passes, where the dynamic range is more bounded. The VAE mean is bounded by $\tanh(\cdot) \times 10$ to prevent unbounded latent values, and we apply `torch.nan_to_num` followed by `tanh clamp` to the decoder output to ensure finite waveform values.

Frozen vs. trainable parameters. Table 9 reports parameter counts. The partial freezing strategy preserves the pretrained codec’s acoustic feature extraction (encoder blocks 0–3) and waveform synthesis (decoder blocks 1–4) while allowing the latent bottleneck and temporal resampling modules to adapt to the target $d \times f$ configuration.

Three-stage curriculum settings. The post-hoc compressor (Table 3 in main text) uses slightly different settings from the grid training, as it compresses a *pretrained* codec rather than retraining from scratch:

F Training Dynamics

Figures 6–8 show the full training dynamics of all 16 retrained grid configurations, including the PESQ trajectories in Figure 7. All models were trained for 40 epochs with the same optimizer, learning-rate schedule, 10,000 steps per epoch, and GAN warmup at epoch 20; batch size is 9–10 depending on memory. The discontinuity at epoch

20 marks the transition from reconstruction-only training to adversarial fine-tuning.

Several patterns emerge from these curves:

- **Convergence stability.** All 16 configurations converge without divergence, indicating that the curriculum remains stable even at extreme compression (e.g., 128d@3.125 Hz).
- **GAN transition.** At epoch 20, training losses rise briefly as the discriminator engages. PESQ generally improves after this transition, while SI-SDR can decrease in some settings, reflecting the same perception–distortion trade-off observed in Table 3.
- **Loss floor depends on Hz, not d .** The asymptotic `val_loss` values cluster by column (Hz) rather than by row (dim). For example, all 25 Hz models reach `val_loss` $\approx$ 135–185 regardless of dimension, whereas 3.125 Hz models plateau at `val_loss` $\approx$ 285–305. This is the core evidence for temporal-rate dominance (Section 4).
- **PESQ saturation.** 25 Hz and most 12.5 Hz models saturate near PESQ $\approx$ 4.0–4.5 (MOS-equivalent), while many 3.125 Hz configurations are capped near PESQ $\approx$ 2.3–2.8, suggesting a strong perceptual ceiling imposed by temporal undersampling.

G Band-wise Ablations

To attribute the Stage 2 gain (Section 4.3) to specific components of the band-wise loss, we retrain the 256d@12.5 Hz checkpoint for 20 additional epochs with one band-wise component disabled at a time. The ablations are: `no_bands` (full-band only, no LF/MF/HF decomposition), `no_overlap` (hard-split bands, no overlapping windows), `no_freq_weight` (uniform per-mel-bin weights), `no_temporal_grad` (drop $\mathcal{L}_{\text{grad}}^b$), and `no_2d_grad` (drop the HF joint time-frequency gradient $\mathcal{L}_{\text{2D}}^{\text{HF}}$). All ablations resume from the converged Stage 3 checkpoint of the 256d@12.5 Hz grid model so that absolute deltas reflect the marginal contribution of each component on top of a fully-trained model rather than re-running the entire curriculum.

Reading the table. The full bandwise loss is strongest overall, especially on speech and music, but a few sound-domain metrics are comparable or slightly better under ablation (e.g.,

Parameter	Value
<i>Architecture</i>	
Base codec	facebook/sam-audio-large (DAC-VAE)
Frozen blocks	encoder.block[0–3], decoder.model[1–4]
Trainable blocks	encoder.block[4–6], quantizer, decoder.model[0]
Temporal resampling	LatentDown/Up (Conv1d, kernel=4, stride=2)
VAE mean bound	$\tanh(\cdot) \times 10$
Activation	StableSnake1d ($\alpha = \text{softplus}(\alpha_{\text{raw}}) + \epsilon$)
<i>Optimization</i>	
Optimizer	AdamW ($\beta_1 = 0.8, \beta_2 = 0.99, \text{wd} = 10^{-4}$)
Learning rate	4×10^{-5} (generator and discriminator)
LR schedule	Exponential decay ($\gamma = 0.999996$ per step)
Batch size	10 per GPU (adjusted to 9 for memory)
Epochs	40
Max steps per epoch	10,000
Gradient checkpointing	Enabled
<i>GAN Training</i>	
Discriminator	MPD $\times 5$ + MRD $\times 3$
GAN warmup	20 epochs (reconstruction-only before GAN)
GAN loss	Least-squares (LSGAN)
Discriminator AMP	fp16 (codec path stays fp32)
<i>Data</i>	
Sample rate	48 kHz
Segment length	5.12s (245,760 samples)
Audio hours per epoch	$\sim 142\text{h}$ (10K steps $\times$ 10 batch $\times$ 5.12s)
Total audio hours	$\sim 5,680\text{h}$ (40 epochs)
<i>Infrastructure</i>	
GPUs per model	1 (NVIDIA B200, 192 GB)
Parallel models	2 per GPU (memory permits)
Training time per model	~ 27 hours

Table 8: Hyperparameters for the 4×4 grid training. Configurations share the same training recipe except for latent dimension d , temporal rate f , and memory-driven batch-size adjustment.

Component	Params	Trainable	Frozen
Encoder blocks 0–3 (frozen)	5.2M	0	5.2M
Encoder blocks 4–6 (trainable)	1.6M	1.6M	0
Latent down/up (n steps)	0.3–0.9M	0.3–0.9M	0
VAE in/out proj (d -dependent)	0.1–1.0M	0.1–1.0M	0
Decoder block 0 (trainable)	0.8M	0.8M	0
Decoder blocks 1–4 (frozen)	18.5M	0	18.5M
Discriminator (MPD+MRD)	28.3M	28.3M	0
Total	$\sim 55\text{M}$	$\sim 31\text{--}32\text{M}$	$\sim 24\text{M}$

Table 9: Parameter counts for different grid configurations.

sound SI-SDR for `no_bands` and sound PESQ for `no_temporal_grad`). The largest degradations cluster around `no_overlap` (SI-SDR drops 1.7dB on speech and 7.5dB on music) and `no_freq_weight` (SI-SDR drops 8.6dB on music; sound SI-SDR turns negative). `no_temporal_grad` and `no_2d_grad` cause smaller but still measurable drops, primarily on music where transient detail is most demanding. Overall, no single ablation improves consistently across

metrics, and the most load-bearing components appear to be overlap and frequency weighting.

Caveat. Because the original Stage-1 checkpoints from prior work were unrecoverable (storage corruption), the ablations resume from the Stage-3 grid checkpoint rather than from a fresh Stage-1 starting point. Absolute numbers are therefore smaller than they would be in a from-scratch curriculum re-run; the relative ordering across ablations is the load-bearing comparison.

H Frequency Weighting Slope Sensitivity

I Multi-seed Variance Check

As a targeted robustness check, three-seed experiments show cross-seed PESQ variance of ± 0.04 at 256d@12.5 Hz and ± 0.06 at 512d@6.25 Hz. These variances are much smaller than the representative inter-configuration gap of 0.49 PESQ points, suggesting that the main reconstruction trends are not seed artifacts.

Stage	Epochs	LR	Batch	Key addition
1 (full-band)	50	3×10^{-5}	8×8 GPU	Mel + STFT + KL + GAN (after 5-ep warmup)
2 (band-wise)	until conv.	3×10^{-5}	8×8 GPU	+ LF/MF/HF band losses
3 (adversarial)	5 warmup + cont.	3×10^{-5}	5×8 GPU	+ Hinge GAN refinement

Table 10: Per-stage training configuration for the post-hoc compressor (256d, 12.5 Hz).

Training Loss Curves: 4×4 Dim $\times$ Hz Grid

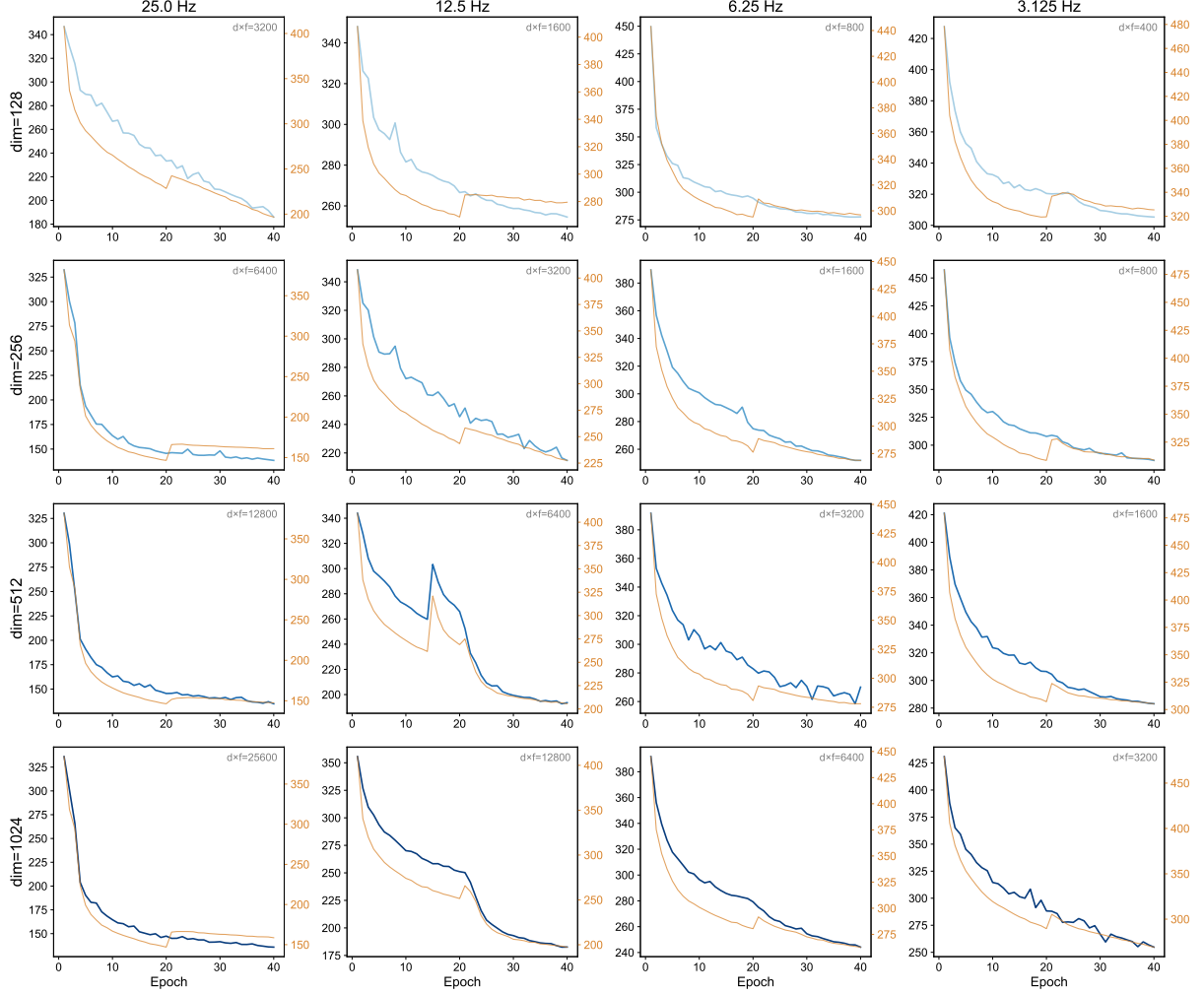


Figure 6: Validation loss curves across the 4×4 dim $\times$ Hz grid. Blue: val_loss; orange (right axis): train_total_g. The discontinuity at epoch 20 corresponds to GAN warmup onset.

J Per-band Error and Capacity Regression

Figure 9 shows the full 4×4 grid of per-band MSE for the speech domain, on a log scale. The dominant pattern is that LF (20–512 Hz) and MF (512–2048 Hz) errors account for most of the measurable temporal-rate-induced increase, while HF MSE (2048–8000 Hz) remains comparatively flat across the grid.

Figure 10 fixes $d = 256$ and sweeps temporal rate, showing LF/MF/HF trends across

speech/sound/music. The qualitative pattern (LF degrades fastest, HF most stable) is preserved across modalities.

Figure 11 reveals the secondary observation that within each band, increasing d partially mitigates the temporal-rate damage but does not eliminate it within the tested range, supporting the view that capacity reallocation along the dimension axis has diminishing returns once temporal sampling is the bottleneck.

Validation PESQ: 4×4 Dim × Hz Grid

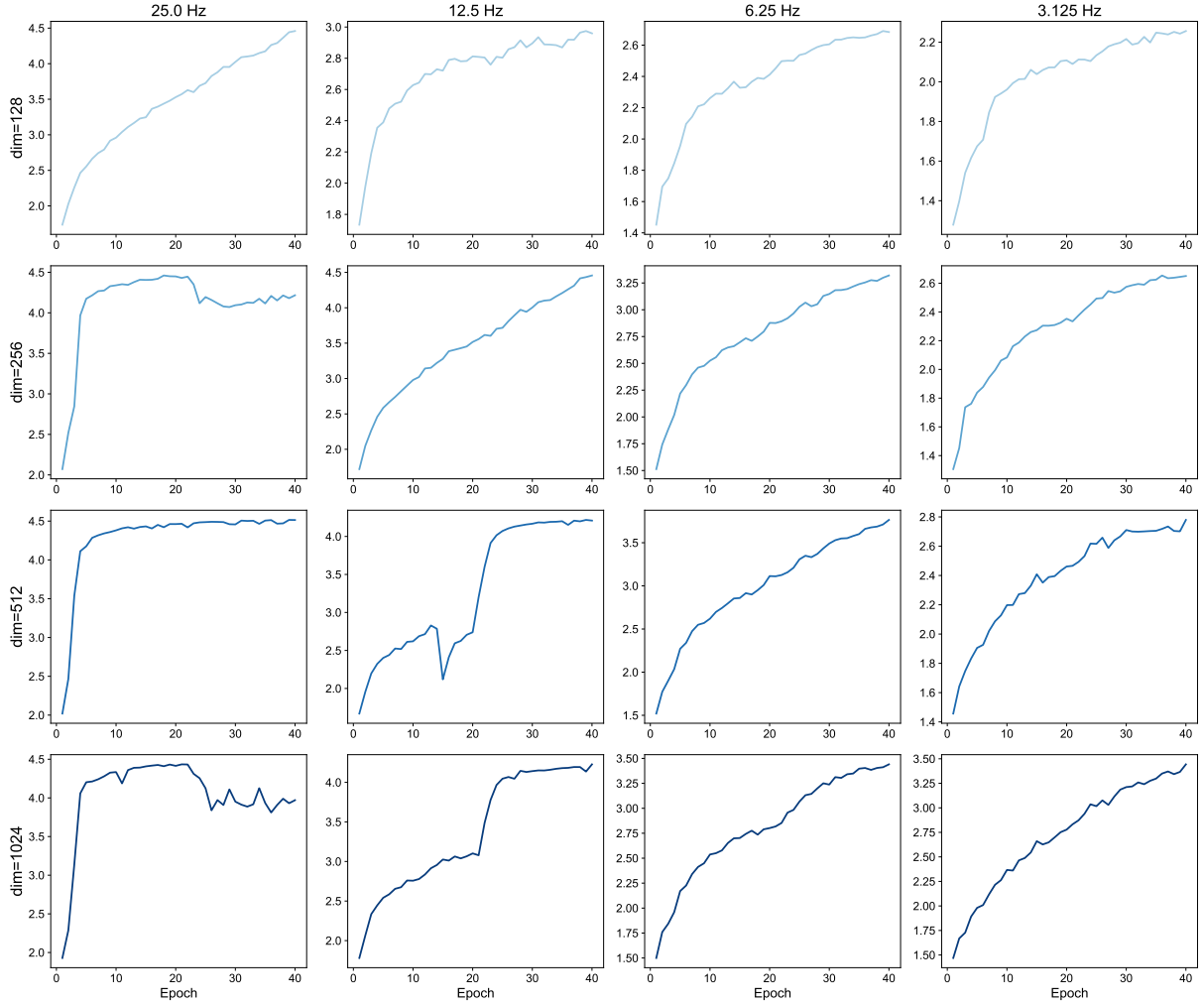


Figure 7: Validation PESQ across the 4×4 dim×Hz grid. PESQ generally improves after GAN warmup (epoch 20) and saturates by epoch 30–35. The asymptotic ceiling is largely bounded by temporal rate.

Variant	Speech			Sound			Music		
	SI-SDR↑	PESQ↑	MSD↓	SI-SDR↑	PESQ↑	MSD↓	SI-SDR↑	PESQ↑	MSD↓
Full (baseline)	12.07	4.46	0.16	0.26	4.20	0.25	23.00	4.43	0.20
no_bands	11.73	4.37	0.20	0.35	4.10	0.30	17.06	4.26	0.41
no_overlap	10.33	4.31	0.21	−0.18	3.99	0.32	15.53	4.11	0.40
no_freq_weight	10.49	4.40	0.19	−0.79	4.16	0.28	14.37	4.28	0.34
no_temporal_grad	10.59	4.37	0.19	−0.58	4.23	0.28	14.57	4.29	0.36
no_2d_grad	10.73	4.41	0.19	−0.59	4.20	0.28	14.52	4.29	0.35

Table 11: Band-wise component ablation (256d@12.5 Hz, 50 utterances per modality). The first row is the full bandwise baseline (the 256d@12.5 Hz cell from Table 1); each subsequent row disables one component. Within each *column*, **bold** marks the worst (most degraded) ablated value: lowest SI-SDR/PESQ or highest MSD. Across columns, the worst-performer pattern concentrates on `no_overlap` and `no_freq_weight`, indicating that overlapping band windows and frequency-axis weighting are the most load-bearing components.

Capacity regression details. Figure 12 shows the log-linear regression discussed in §??, fit to all 16 retrained grid cells. Each panel shows PESQ, SI-SDR, or MSD plotted against $d \cdot f$ (left column)

and $d \cdot f^2$ (right column), with separate lines per modality. R^2 values for $d \cdot f^2$ exceed those for $d \cdot f$ in every modality and metric, providing quantitative support for the empirical scaling hypothesis.

Validation SI-SDR (dB): 4×4 Dim $\times$ Hz Grid

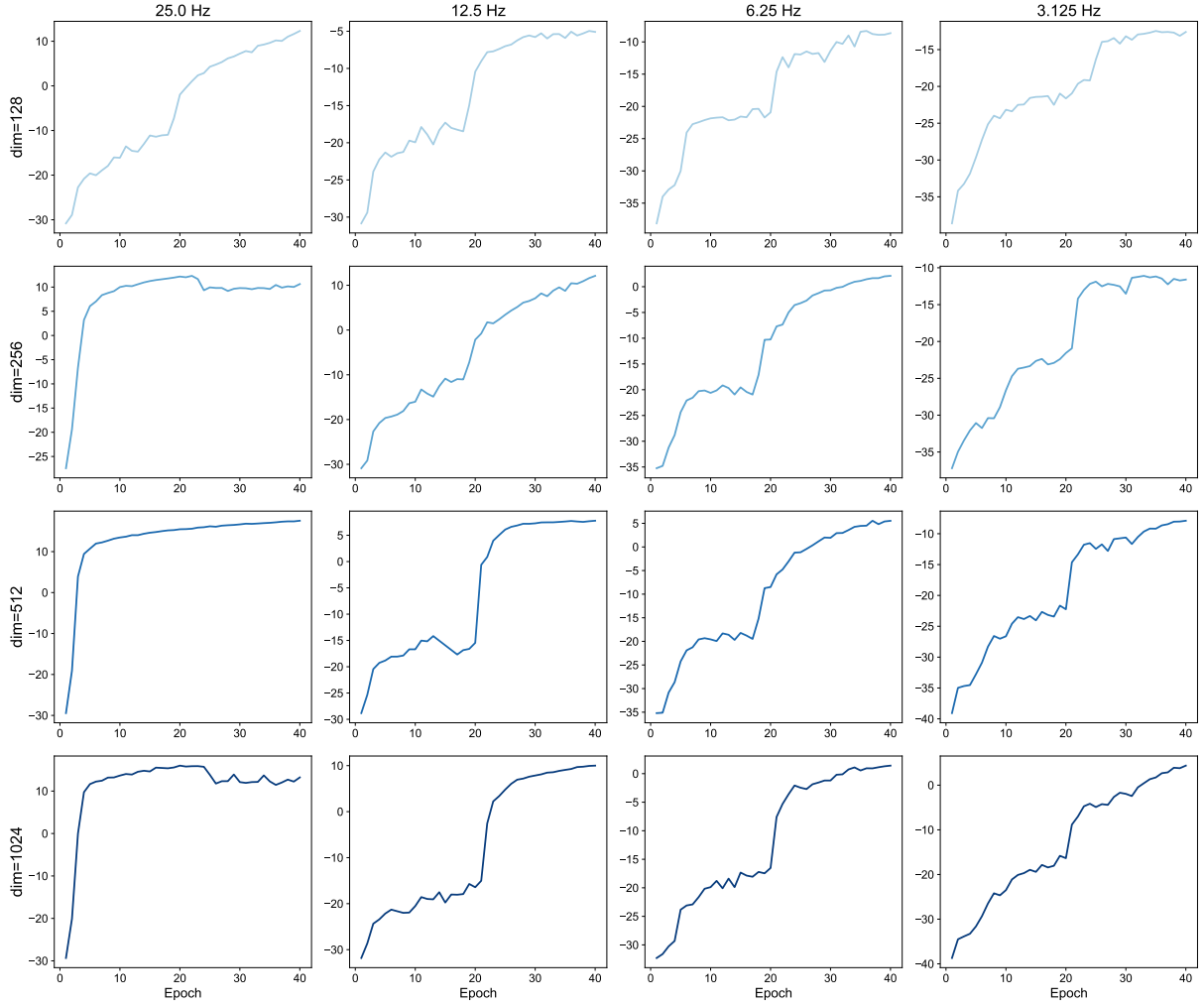


Figure 8: Validation SI-SDR across the 4×4 dim $\times$ Hz grid. SI-SDR shows the widest spread across Hz, from $\sim +17$ dB (25 Hz) to ~ -13 dB (3.125 Hz). Dimension variations within a fixed Hz contribute comparatively little.

Config	s_{HF}	SI-SDR	PESQ	STOI	UTMOS	MSD
Uniform	1.0	6.64	4.31	0.926	3.77	0.23
Default	2.0	5.44	3.11	0.948	3.53	2.29
Aggressive	3.0	10.35	4.30	0.906	3.86	0.28

Table 12: Sensitivity to the HF weighting slope s_{HF} . Slope changes affect the PESQ/SI-SDR/MSD trade-off, so gains should not be attributed solely to the slope value.

Config	d	f (Hz)	SI-SDR $\uparrow$	UTMOS $\uparrow$	MSD $\downarrow$
128d, 25 Hz	128	25.0	23.94	1.33	0.19
256d, 12.5 Hz	256	12.5	23.00	1.34	0.20
512d, 6.25 Hz	512	6.25	17.61	1.33	0.25
1024d, 3.125 Hz	1024	3.125	8.03	1.33	0.35

Table 13: Constant-bandwidth trade-off on the music domain.

K Domain-level Sensitivity

Figures 13–15 visualize how compression sensitivity varies by modality across the 4×4 grid, with the fixed-256d rate sweep shown in Figure 14. The ordering depends on the metric: sound effects degrade most strongly in diagonal PESQ, music shows large SI-SDR drops and qualitative loss of harmonic detail, and speech remains comparatively

robust in PESQ because much linguistically relevant information is concentrated in slowly-varying formant trajectories.

L Music-domain Diagonal Results

Music exhibits the same phase transition along the diagonal (constant $d \times f \approx 3200$): SI-SDR is 23.94 dB at 128d@25 Hz, 23.00 dB at 256d@12.5 Hz, drops to 17.61 dB at 512d@6.25 Hz, and falls further to 8.03 dB

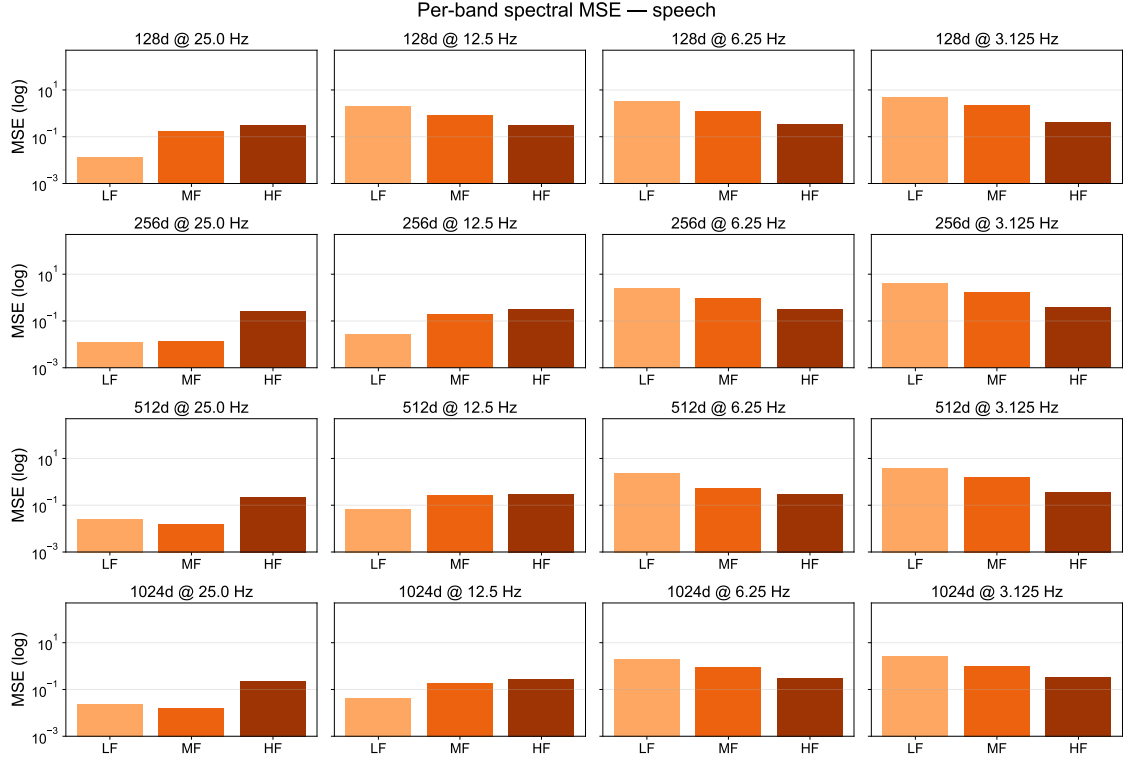


Figure 9: Per-band spectral MSE across the 4×4 dim $\times$ Hz grid (speech, log scale). LF/MF rise with lower temporal rates; HF remains comparatively flat under this metric.

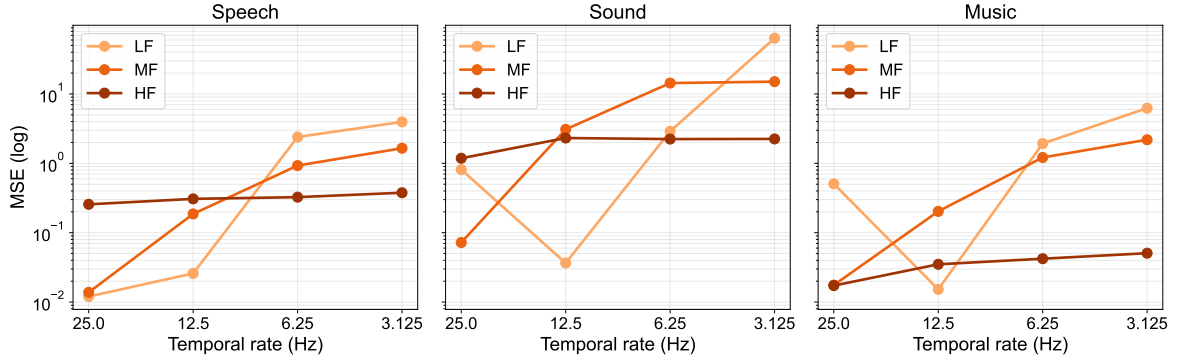


Figure 10: Per-band MSE vs. temporal rate, fixed $d = 256$. Rows: bands; columns: modality.

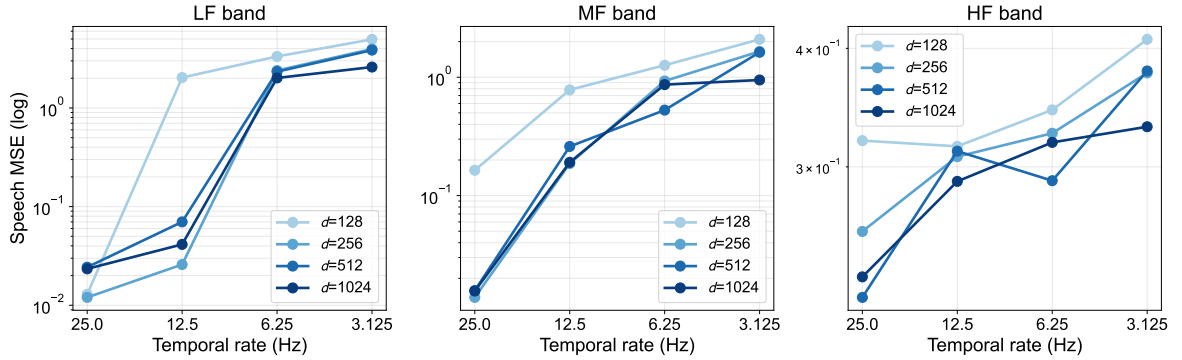


Figure 11: Speech LF/MF/HF MSE vs. temporal rate, with $d \in \{128, 256, 512, 1024\}$ overlaid.

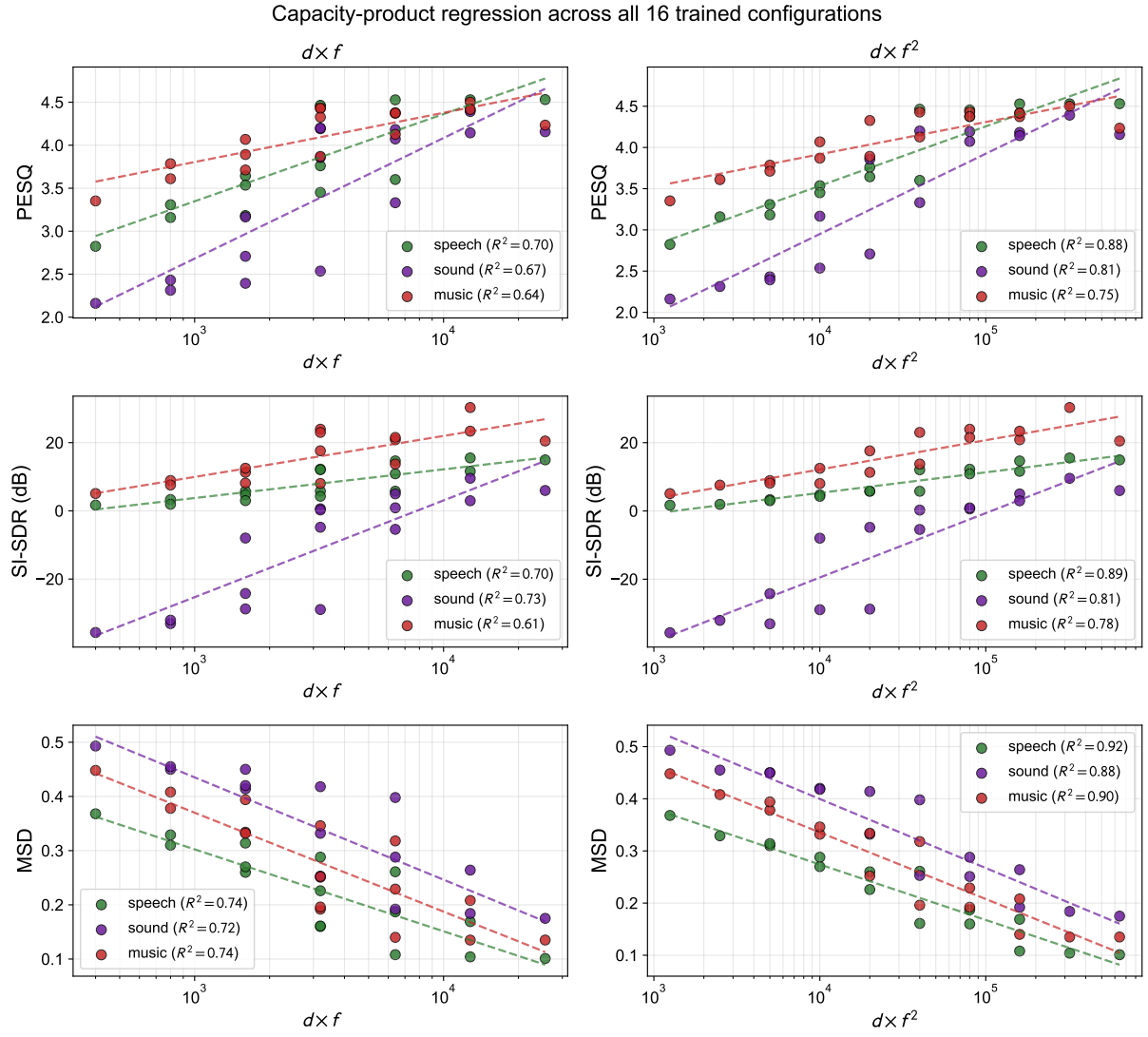


Figure 12: Log-linear regression of reconstruction quality on $d \cdot f$ (left) vs. $d \cdot f^2$ (right) across all 16 retrained configurations. R^2 is shown in the legend; the $d \cdot f^2$ predictor explains more variance for every metric and modality in this grid.

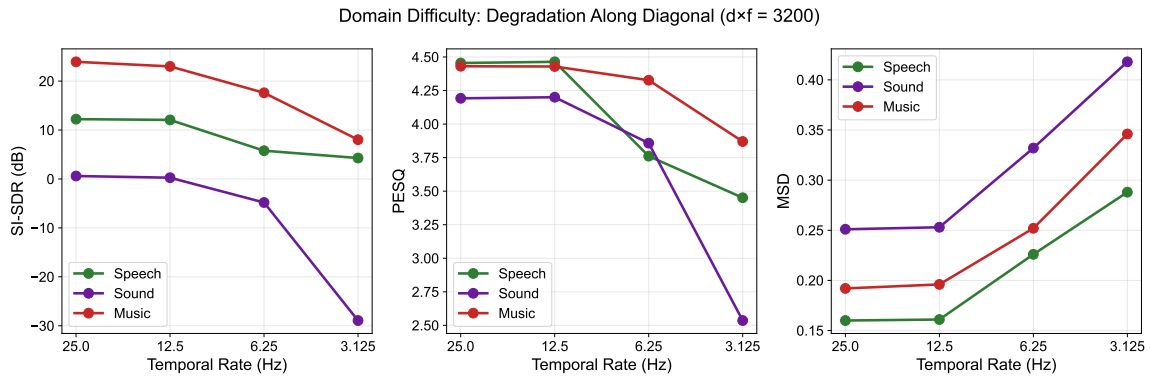


Figure 13: Diagonal configurations ($d \cdot f \approx 3200$): metric-dependent collapse across domains. Sound shows the largest PESQ/SI-SDR degradation, while speech and music are comparatively more robust in PESQ.

at 1024d@3.125 Hz. This shows substantial SI-SDR degradation for music, although cross-

domain difficulty remains metric-dependent rather than a single speech–sound–music ordering.

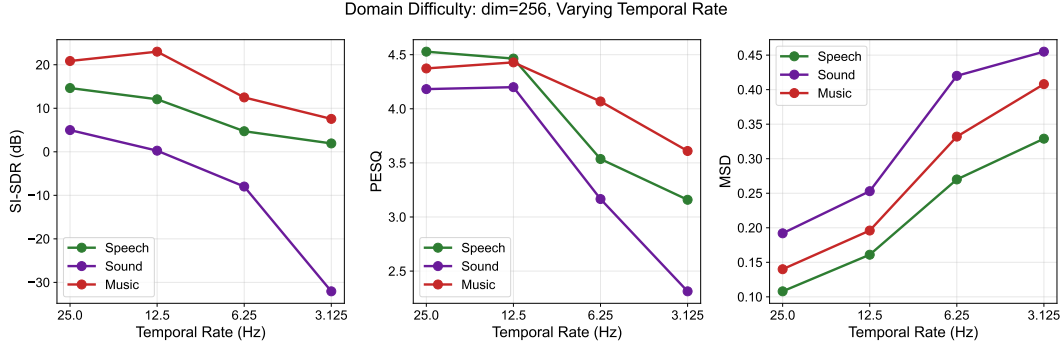


Figure 14: Fixed $d = 256$ slice across temporal rates: PESQ degradation is monotonic for all domains, with the largest loss appearing in the sound domain.

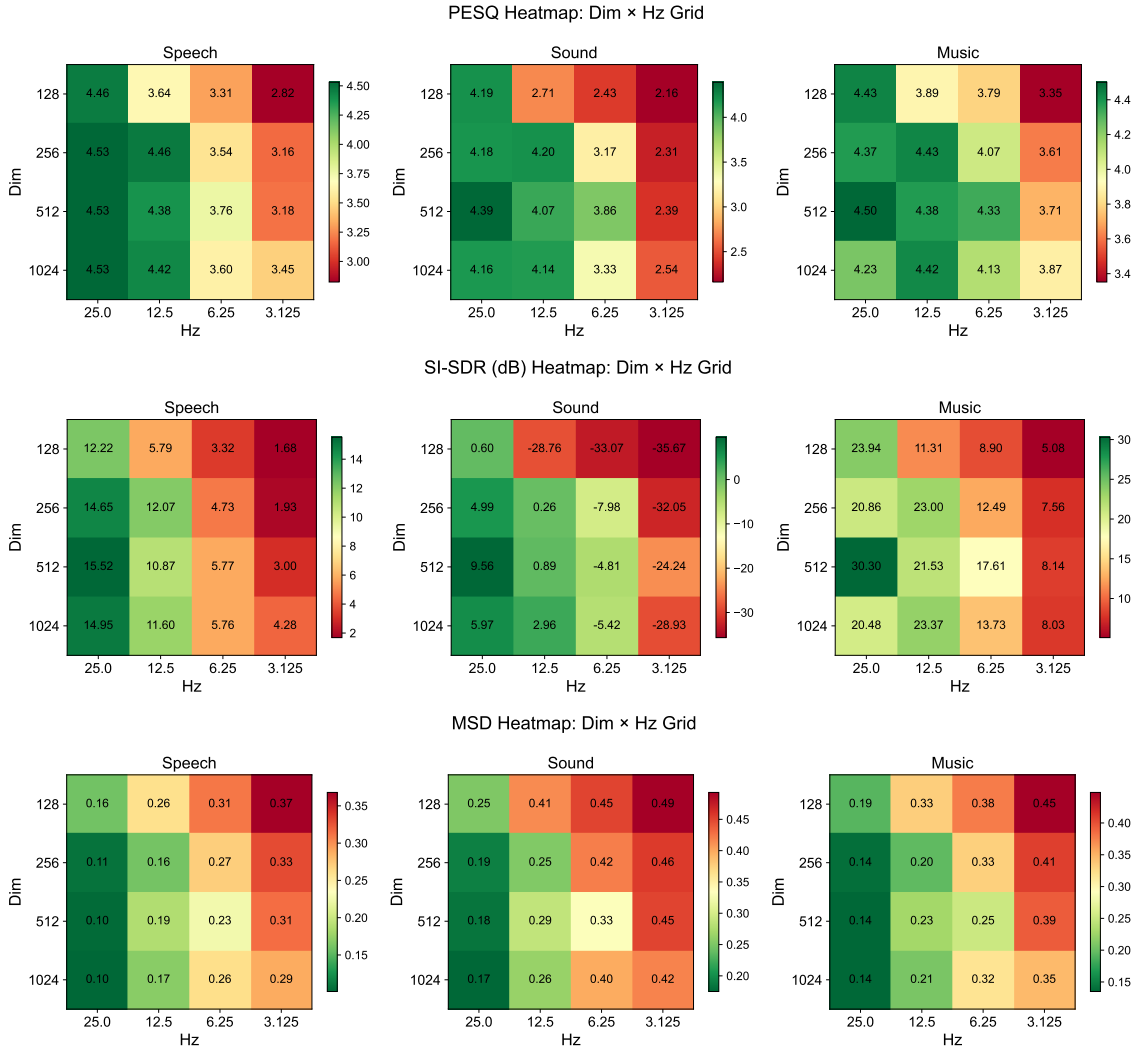


Figure 15: Per-domain heatmaps across the full 4×4 grid, one row per metric: PESQ ($\uparrow$), SI-SDR ($\uparrow$, dB), MSD ($\downarrow$). Within each row, the three panels show speech, sound, and music. The heatmaps show metric-dependent sensitivity: sound drops sharply in PESQ/SI-SDR, while music shows strong SI-SDR degradation and broader temporal-detail loss.

M Concatenation Baseline Diagnostics

Figure 16 shows per-utterance metrics for each of the three conditions evaluated in Table 4. Original-

25 Hz and Concat-256d@12.5 points coincide on the $y = x$ line for every metric, as expected from an information-preserving reversible 2 : 1

frame reshape evaluated under the original decoder path. The Learned-256d@12.5 cloud sits below the $y = x$ line on SI-SDR but above on PESQ and UTMOS, and far below on MSD, providing per-utterance evidence that learned compression can redistribute capacity toward perceptually salient structure, subject to the decoding-path caveat noted in the main body.

Figures 17 show mel-spectrogram reconstructions for two representative utterances. The original DAC-VAE and the Concat path are visually indistinguishable from the ground truth, while the Learned model produces a slightly smoothed but still natural mel envelope, with cleaner mid-frequency formant tracks at the cost of mild high-frequency blurring.

N Latent Information Diagnostics

We measure per-dimension entropy and adjacent-frame mutual information using the KSG estimator over 10,000 validation frame pairs. Per-dimension entropy remains stable at 6.18–6.47 bits across configurations, but adjacent-frame MI rises sharply from 0.49 bits at 25 Hz to 0.84 bits at 3.125 Hz. Reconstruction quality begins to degrade sharply once adjacent-frame MI exceeds roughly 0.55 bits (Figure 18a). This is consistent with a capacity-allocation issue: at high compression, adjacent frames become more redundant while less capacity remains for novel temporal detail. For comparison, Figure 18b shows that all four naive downsampling baselines, average pooling, strided selection, max pooling, and nearest-neighbor upsampling at 12.5 Hz, produce near-floor PESQ (~ 1.2 – 1.4) with negative SI-SDR, indicating that learned compression is necessary for usable quality in this setting.

O Qualitative Mel Spectrograms

The 128d@25 Hz and 256d@12.5 Hz reconstructions preserve most visible harmonic structure below 4 kHz and retain energy in the 4–8 kHz region (Figure 19). At 512d@6.25 Hz, harmonics above roughly 3 kHz become more blurred, while the 1024d@3.125 Hz example loses much of the fine harmonic structure. The stage comparison (Figure 20) suggests a qualitative progression: Stage 1 preserves fundamentals, Stage 2 restores clearer 1–4 kHz structure, and Stage 3 adds finer broadband texture in this example.

P ASR Experiment Details

We provide implementation details for the ASR experiments in Section 4.4.3. For each codec configuration we freeze the trained encoder and project its latents into a frozen Qwen3.5-4B language model via a 4-layer causal transformer adapter. The adapter is trained with cross-entropy loss on LibriSpeech, GigaSpeech, Voxpopuli, and Multilingual LibriSpeech (MLS) ASR transcription tokens, totaling approximately 56,000 hours of audio. Table 15 summarizes the adapter training configuration. At inference time we use greedy decoding with batch size 16. We evaluate on LibriSpeech `test-clean` and `test-other`; which are lowercased and stripped of non-alphanumeric characters before scoring, and CER is additionally computed on space-removed strings. Table 14 shows full results with bootstrap standard deviations using 10,000 utterance-level resamples with replacement.

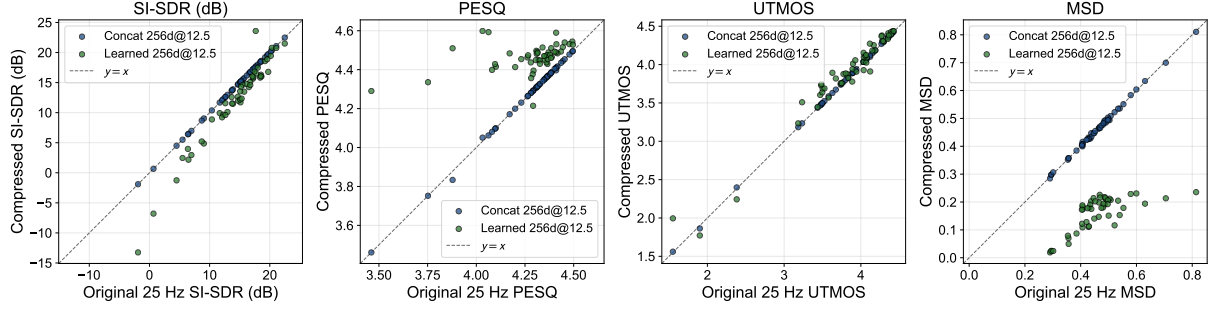


Figure 16: Per-utterance scatter (50 speech files) of compressed-vs-original metrics. Concat (blue) lies on the $y = x$ identity line; Learned (green) gains PESQ/UTMOS/MSD at the cost of small SI-SDR loss.

Table 14: Full ASR results corresponding to Table 6, with bootstrap standard deviations (10,000 utterance-level resamples with replacement). Reported as point estimate $\pm$ std; lower is better.

Dim	Rate (Hz)	Libri. Clean		Libri. Other	
		WER	CER	WER	CER
128	3.125	8.40 $\pm$ 0.20	4.54 $\pm$ 0.12	26.30 $\pm$ 0.55	16.98 $\pm$ 0.40
	6.25	4.95 $\pm$ 0.15	2.30 $\pm$ 0.08	17.92 $\pm$ 0.73	11.17 $\pm$ 0.62
	12.5	4.22 $\pm$ 0.13	1.92 $\pm$ 0.07	14.07 $\pm$ 0.29	8.29 $\pm$ 0.19
	25.0	4.25 $\pm$ 0.14	1.90 $\pm$ 0.07	14.15 $\pm$ 0.30	8.37 $\pm$ 0.19
256	3.125	7.60 $\pm$ 0.19	3.94 $\pm$ 0.11	23.47 $\pm$ 0.42	15.02 $\pm$ 0.35
	6.25	4.90 $\pm$ 0.15	2.28 $\pm$ 0.07	16.94 $\pm$ 0.44	10.31 $\pm$ 0.38
	12.5	4.48 $\pm$ 0.14	2.10 $\pm$ 0.07	15.00 $\pm$ 0.31	8.83 $\pm$ 0.19
	25.0	4.27 $\pm$ 0.13	1.95 $\pm$ 0.07	14.77 $\pm$ 0.31	8.58 $\pm$ 0.19
512	3.125	7.39 $\pm$ 0.19	3.88 $\pm$ 0.10	23.17 $\pm$ 0.37	14.64 $\pm$ 0.25
	6.25	4.59 $\pm$ 0.14	2.12 $\pm$ 0.07	15.77 $\pm$ 0.31	9.34 $\pm$ 0.20
	12.5	4.27 $\pm$ 0.13	1.93 $\pm$ 0.07	14.93 $\pm$ 0.31	8.74 $\pm$ 0.19
	25.0	4.40 $\pm$ 0.14	2.03 $\pm$ 0.07	14.71 $\pm$ 0.30	8.56 $\pm$ 0.18
1024	3.125	7.39 $\pm$ 0.18	3.86 $\pm$ 0.10	23.33 $\pm$ 0.59	14.68 $\pm$ 0.37
	6.25	4.91 $\pm$ 0.15	2.32 $\pm$ 0.08	16.92 $\pm$ 0.33	10.13 $\pm$ 0.21
	12.5	4.36 $\pm$ 0.14	1.98 $\pm$ 0.08	14.59 $\pm$ 0.30	8.54 $\pm$ 0.19
	25.0	4.36 $\pm$ 0.14	2.00 $\pm$ 0.08	15.00 $\pm$ 0.30	8.76 $\pm$ 0.19

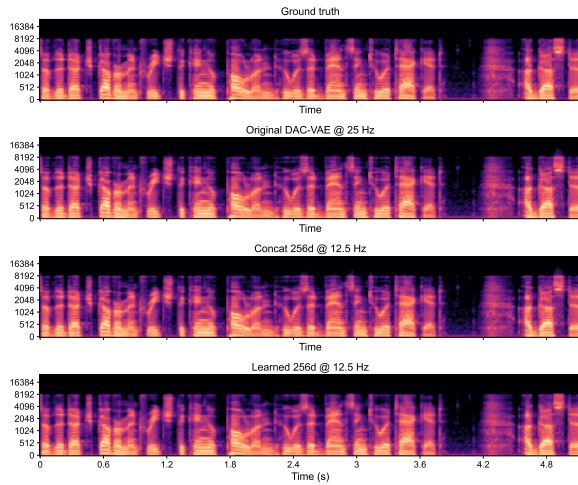


Figure 17: Mel reconstruction example

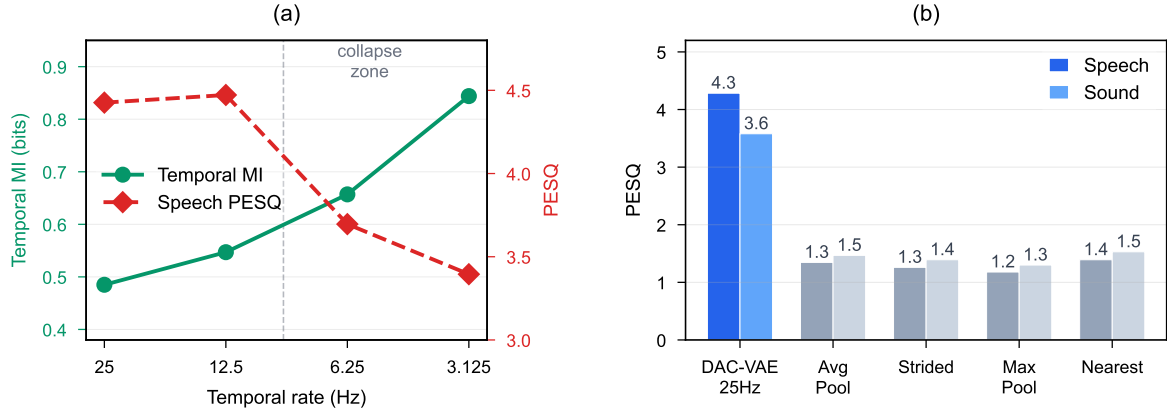


Figure 18: **(a)** Adjacent-frame MI rises as temporal rate decreases, while PESQ drops sharply. **(b)** Naive downsampling methods produce near-floor PESQ in this setup.

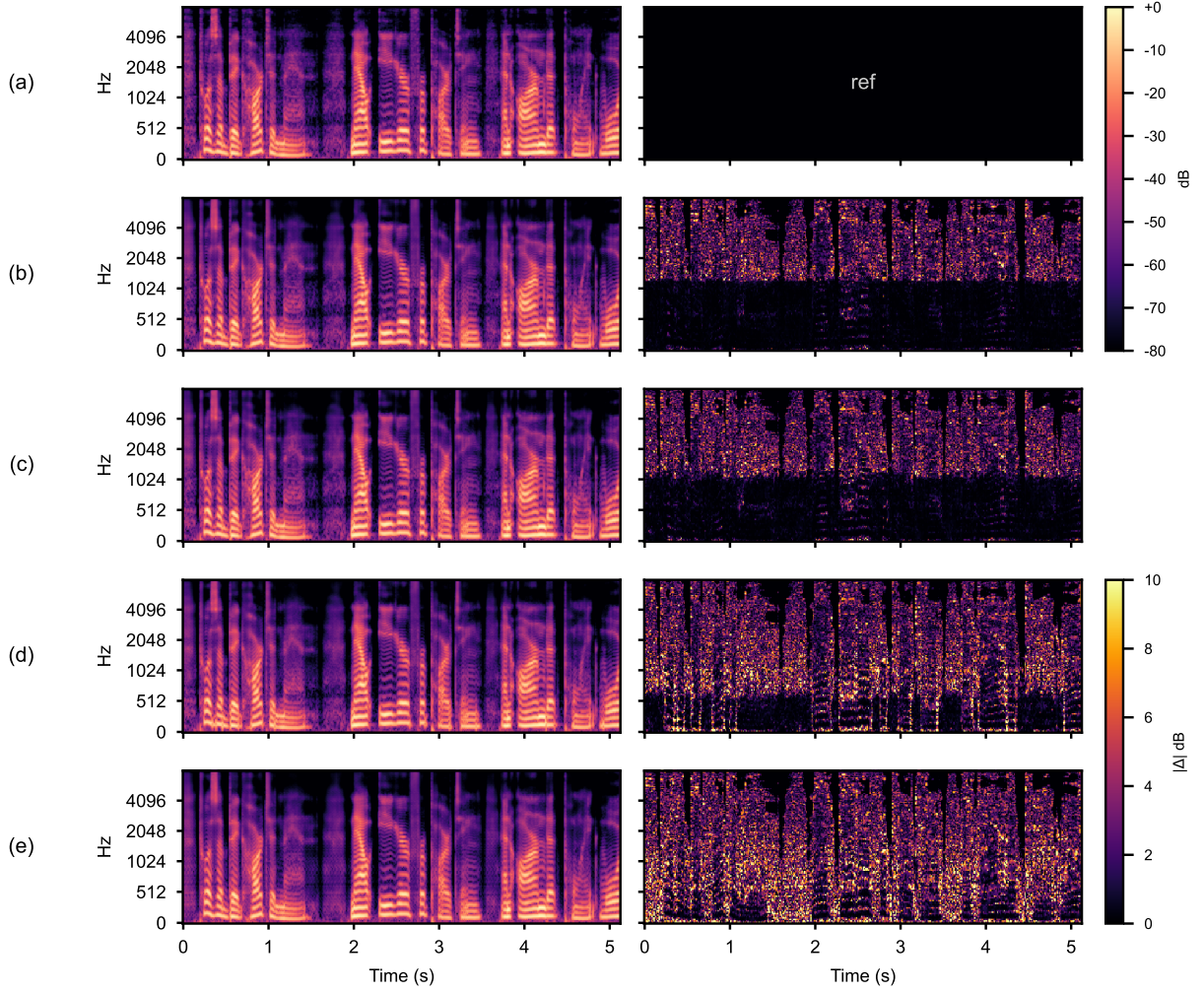


Figure 19: Differences from the ground truth according to the Mel-spectrogram and dimension-temporal rate. **(a)** Ground truth. **(b)** 128d@25 Hz. **(c)** 256d@12.5 Hz. **(d)** 512d@6.25 Hz. **(e)** 1024d@3.125 Hz. Right column represents the difference between ground truth and prediction.

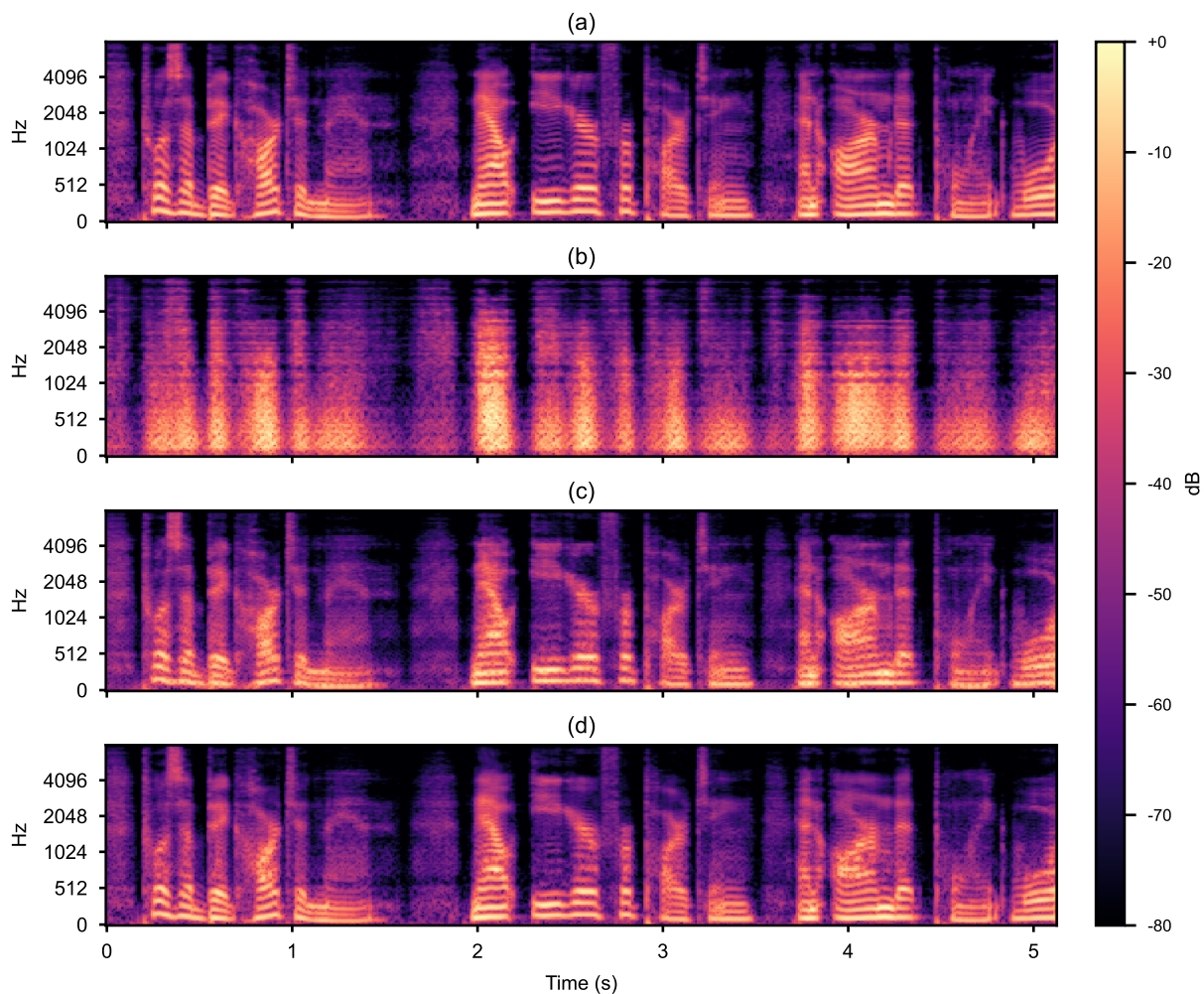


Figure 20: Frequency-detail changes across training stages (256d@12.5 Hz). **(a)** Ground truth. **(b)** Stage 1. **(c)** Stage 2. **(d)** Stage 3. Stage 2/3 recover more visible mid- and high-frequency texture in this example.

Table 15: Adapter-based ASR training hyperparameters.

Parameter	Value
LLM backbone	Qwen3.5-4B (frozen)
Adapter type	4-layer pre-norm causal Transformer
Adapter hidden dim	512
Adapter attention heads	8
Adapter feedforward dim	2048
Dropout	0.0
Trainable params	~18M (adapter only)
Optimizer	AdamW ($\text{lr}=2 \times 10^{-4}$, $\text{wd}=0.01$)
LR schedule	Cosine annealing, 5000-step warmup
Training Steps	50000
Loss	Cross-entropy on ASR transcription tokens
Eval	WER/CER on LibriSpeech test sets